

## PROJECT REPORT – Multi-Game Scratch Collection

### 1. Project Title

Multi-Game Arcade in Scratch

### 2. Introduction

This project is a collection of mini-games created using Scratch. It includes four games:

- Rock Paper Scissors
- Tic Tac Toe
- Flappy Bird
- Dinosaur Game

The goal is to allow players to choose from different games and enjoy multiple types of gameplay inside a single Scratch project.

### 3. Objective

- To create a multi-game system inside Scratch.
- To learn random functions, movement controls, and game logic.
- To design different types of games such as reaction games, jumping games, and simple AI games.
- To practice menu design and switching between games.

### 4. Game Descriptions

#### 1. Rock Paper Scissors

- Player chooses Rock, Paper, or Scissors.
- Computer randomly selects a move.
- Game compares choices and shows Win/Lose/Draw.
- Uses random blocks and condition checking.

#### 2. Tic Tac Toe

- A grid-based logic game.
- Currently under development.
- Uses mouse clicks to place X and O.

- Will include win-checking logic later.

### 3. Flappy Bird

- Bird flies by clicking mouse, pressing Space, W, or Up Arrow.
- Player avoids obstacles.
- Uses gravity, jump controls, and collision detection.

### 4. Dinosaur Game

- Similar to Chrome Dino game.
- Player jumps using mouse click, Space, W, or Up Arrow.
- Avoids obstacles like cacti.
- Uses speed control, movement, and hit detection.

### 5. Features of the Project

- Multiple games inside one Scratch project.
- Easy controls (mouse + keyboard).
- Random logic (Rock Paper Scissors).
- Collision, gravity, and movement scripts.
- Menu-based navigation.
- Fun and beginner-friendly.

### 6. Coding Concepts Used

- Events: Key press, mouse click, broadcasts
- Random blocks: computer choice
- Variables: Game state, score
- Control blocks: Loops, if-else
- Sensing blocks: Touching obstacles
- Motion blocks: Jumping and gravity simulation

### 7. Conclusion

This Scratch project successfully combines multiple games into one platform. It shows understanding of random logic, movement control, game state management, and simple AI behavior. It is creative, interactive, and a great example of beginner-level game development.